Mutagenicity testing in the Salmonella typhimurium assay of phenolic compounds and phenolic fractions obtained from smokehouse smoke condensates.

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Smokehouse smoke, which is used for flavouring meat products, was investigated for its mutagenic activity in the Salmonella typhimurium assay. We were chiefly concerned with the fractions free of polycyclic aromatic hydrocarbons but containing phenol compounds, which are responsible for the preservative and aromatizing properties of the smoke. The most abundantly occurring phenol compounds (phenol, cresols, 2,4-dimethylphenol, brenzcatechine, syringol, eugenol, vanilline and guaiacol) gave negative results when they were tested for mutagenicity at five concentrations up to 5000 micrograms/plate, with and without S-9 mix, using five strains of S. typhimurium. Even when phenol was further investigated in a variety of test conditions, no induction of his+ revertants was observed. When smokehouse smoke was condensed and fractionated the majority of the various phenolic fractions also gave negative results when tested at five concentrations using five strains of S. typhimurium. However there was a slight increase in the number of revertants in a few cases. The presence in the phenolic fractions of very small amounts of mutagenic impurities, the nature of which needs further investigation, cannot be excluded. These results support the further development of non-hazardous smoke-aroma preparations, based on the phenolic components of smokehouse smoke.